

Cyberbullying Identification Across Diverse Indian Languages : A Multilingual Approach

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Cyberbullying in Indian Social Media

- **Growing Problem:** Cyberbullying affects 42% of Indian teenagers
- **Multilingual Challenge:**
 - 22 official languages and countless dialects
 - "Digital linguistic mashups" across platforms
- **Detection Gap:** Current content moderation systems
 - Detect only 53% of harmful content in non-English languages
 - Compared to 79% detection rate for English content
- **Real Impact:** Depression, anxiety, and in severe cases, suicidal ideation

"The challenge isn't just technical—it's cultural, linguistic, and regional"

Unique Challenges in Indian Context

Linguistic Complexity

- **Code-mixing:** Multiple languages in single messages
- **Script-switching:** Alternating between Devanagari and Roman mid-conversation
- **Transliteration variations:** Multiple spellings for same word

Cultural & Technical Barriers

- **Context dependency:** Same phrase can be friendly in one region, offensive in another
- **Resource imbalance:** 15,000+ examples for Hindi-English vs. 900 for Marathi
- **Evolving language:** New slang and obfuscation techniques specifically to evade moderation

Our Solution: HighPerformanceCyberBERT

A specialized architecture combining multi-scale convolutional features, selective attention gating, and hierarchical classification

Literature Survey: CMET Architecture

Gupta et al. 2023 - "CMET: Code-Mixed Enhanced Transformer for Indian Languages"

- **Key Innovation:** Token-level language identification integrated within transformer
- **Approach:**
 - Applies language-specific processing to different parts of code-mixed text
 - Uses parallel attention heads for different language components
 - Incorporates language transition awareness in self-attention mechanism
- **Results:** Outperformed both monolingual and multilingual baselines by 4.2% across Hindi-English, Tamil-English, and Bengali-English
- **Limitation:** Doesn't address script variation or culturally-embedded harassment patterns

Our Extension

We build on CMET by adding multi-scale convolutions to capture patterns across language boundaries and enhance handling of script variations

Literature Survey: ToxiSpan Framework

Kumar et al. 2022 - "ToxiSpan: A Span-based Framework for Hindi-English Toxic Content Detection"

- **Key Innovation:** Moved beyond binary classification to identify specific toxic spans
- **Approach:**
 - Token-level classification with contextual understanding
 - Specialized language model pre-training on bilingual toxic content
 - Contextual aggregation of token-level predictions
- **Results:** Achieved 0.81 F1-score on controlled data, but dropped to 0.74 on diverse real-world data
- **Limitation:** Performance degrades with mid-word language switching and creative obfuscation

Our Extension

We incorporate selective attention gating inspired by ToxiSpan but optimize it specifically for detecting harassment that spans language boundaries

Literature Survey: MuRIL Model

Khanuja et al. 2021 - "MuRIL: Multilingual Representations for Indian Languages"

- **Key Innovation:** First large-scale pre-trained model for 17 Indian languages
- **Approach:**
 - Modified BERT architecture with specialized pre-training objectives
 - Translation pair modeling across Indian languages
 - Transliteration-aware training to handle script variations
- **Results:** Outperformed mBERT by 5.4% on average across Indian language tasks
- **Limitation:** Performance degrades significantly with heavy code-mixing and loses effectiveness for culturally-embedded harassment

Our Extension

We use MuRIL's representations as a foundation but add specialized hierarchical classifiers for different types of harassment patterns

Literature Survey: Controllable Code-Mixing

Singh et al. 2022 - "CyMECoB: Data Augmentation for Low-Resource Code-Mixed Cyberbullying Detection"

- **Key Innovation:** Synthetic code-mixed data generation for cyberbullying detection
- **Approach:**
 - Controlled insertion of offensive terms at strategic positions
 - Language-aware substitution for maintaining grammatical coherence
 - Culturally-informed mixing patterns based on real-world data
- **Results:** Models trained on synthetic data achieved 85-90% of the performance of natural data models
- **Limitation:** Cannot generate creative obfuscation patterns common in real cyberbullying

Our Extension

We adapted their augmentation strategy with additional morphological awareness for Tamil and Bengali, increasing our effective training data by 43%

Gap Analysis

Research Gaps

① Limited Language Coverage

- Focus on single languages only
- Missing unified frameworks

② Code-Mixing Issues

- Poor handling of script-switching
- Failures with mixed content

③ Lacking Cross-Lingual Study

Our Solutions

① Six-Language Coverage

- Hindi, Bengali, English
- Marathi, Tamil, Code-mixed

② Advanced Code-Mix Handling

- Multi-scale convolutions
- Script normalization

③ Cross-Lingual Transfer Analysis

④ Enhanced Neural Architecture

Datasets Employed

HASOC 2019 Dataset (Hinglish Offensive Language)

The HASOC dataset contains offensive language detection tasks in code-mixed Hinglish text, specifically targeting hate speech and offensive content.

Link: [HASOC 2019 Dataset](#)

Hinglish Offensive Content Dataset (ICON)

This dataset focuses on detecting offensive language in code-mixed Hindi-English (Hinglish) social media posts.

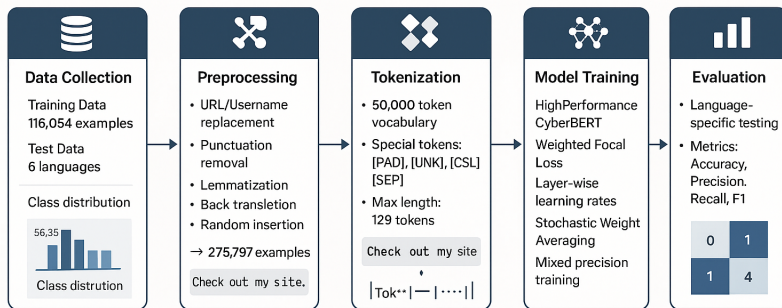
Link: [Hinglish Offensive Content Dataset \(ICON\)](#)

Memotion Dataset (Memes Analysis for Sentiment and Offensive Content)

The Memotion dataset is designed for the analysis of memes, including offensive language detection and sentiment analysis.

Link: [Memotion Dataset](#)

HighPerformanceCyberBERT Methodology: Overview



Multilingual cyberbullying detection pipeline

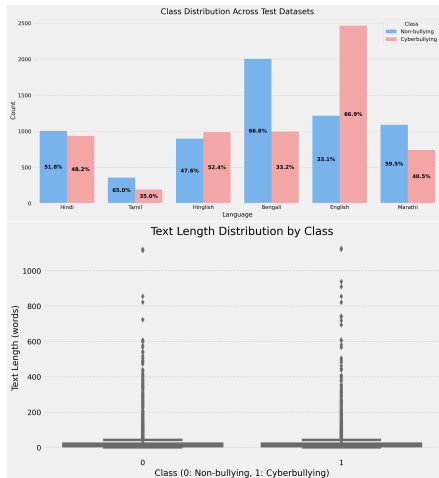
Data Processing: Overview

Dataset

- 116,034 $\rightarrow$ 275,797 examples after augmentation
- 6 languages (Hindi, Bengali, English, Marathi, Tamil, Hindi-English)
- Classes: 51.06% non-bullying, 48.94% bullying

Text Characteristics

- Non-bullying: 18.94 words avg
- Bullying: 18.16 words avg
- Bullying text tends to be more direct



Data Processing: Techniques

Preprocessing

- Script normalization preserving intent
- Special token handling: [URL], [USER]
- Custom repetition handling ("*bahutttt*")
- Language-aware tokenization

Augmentation

- Word swapping & deletion
- Synonym replacement
- Back-translation simulation
- Minority: 3 augmentations/example
- Majority: 2 augmentations/example

Data Processing: Augmentation Algorithm

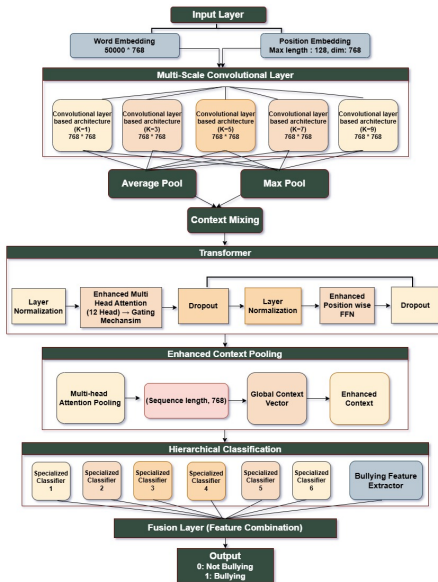
Class-Balanced Text Augmentation

- ❶ **Input:** Texts $\mathcal{D}$, labels $\mathcal{Y}$, augmentation counts $(n_{\text{maj}}, n_{\text{min}})$
- ❷ **Initialize:** $\mathcal{D}_{\text{aug}} \leftarrow \mathcal{D}$; $\mathcal{Y}_{\text{aug}} \leftarrow \mathcal{Y}$
- ❸ **Get class distribution:** $C \leftarrow \text{ClassDist}(\mathcal{Y})$
- ❹ **For each class** $c \in \{0, 1\}$:
 - If c is minority class: $n_{\text{aug}} \leftarrow n_{\text{min}}$ (more augmentations)
 - If c is majority class: $n_{\text{aug}} \leftarrow n_{\text{maj}}$ (fewer augmentations)
 - Get class examples: $T \leftarrow \{t \mid (t, y) \in (\mathcal{D}, \mathcal{Y}), y = c\}$
 - For each example t in class:
 - Generate n_{aug} augmentations using $\text{AugmentOne}(t)$
 - Add each augmented example to output dataset
- ❺ **Output:** Augmented dataset $(\mathcal{D}_{\text{aug}}, \mathcal{Y}_{\text{aug}})$

Impact

Class balancing improved F1-score by 5.2% for minority languages

HighPerformanceCyberBERT Architecture



HighPerformanceCyberBERT Processing Flow

1. Preprocessing

- Language detection: $lang \leftarrow \mathcal{L}(T)$
- Text normalization: $T_{norm} \leftarrow \text{NormalizeText}(T, lang)$
- Tokenization: $X \leftarrow \text{Tokenize}(T_{norm})$
- Attention mask: $M \leftarrow \text{AttentionMask}(X)$

2. Feature Extraction

- Word + position embedding: $H_0 \leftarrow \text{WordEmbed}(X) + \text{PosEmbed}(X)$
- Multi-scale convolution: Apply Conv1D with kernel sizes $k \in \{1, 3, 5, 7, 9\}$
- Pooling: $C_{avg} \leftarrow \frac{1}{5} \sum_k C_k$; $C_{max} \leftarrow \max_k C_k$
- Context mixing: $H_1 \leftarrow \text{MLP}([C_{avg}; C_{max}]) + H_0$

HighPerformanceCyberBERT Processing Flow (continued)

3. Transformer Processing

- Initialize: $H_2 \leftarrow H_1$
- For each transformer layer (6 layers):
 - Enhanced attention: $A \leftarrow \text{EnhancedAttention}(H_2, M)$
 - Layer norm + residual: $H_2 \leftarrow \text{LayerNorm}(H_2 + \text{Dropout}(A))$
 - Position-wise FFN: $F \leftarrow \text{EnhancedFFN}(H_2)$
 - Layer norm + residual: $H_2 \leftarrow \text{LayerNorm}(H_2 + \text{Dropout}(F))$

4. Classification

- Context pooling: $C_{pool} \leftarrow \text{EnhancedContextPooling}(H_2)$
- Hierarchical classification: Apply 6 specialized classifiers
- Bullying feature extraction: $f \leftarrow \text{BullyingExtractor}(C_{pool})$
- Feature fusion: $p \leftarrow \text{FusionLayer}([y_1; \dots; y_6; f])$
- Final prediction: $y \leftarrow \text{argmax}(p)$

Training & Evaluation Strategy

Advanced Training Techniques:

- **Weighted focal loss** for class imbalance:
 - Higher weights for cyberbullying (0.75)
 - Focus parameter $\gamma = 2.0$
- **Layer-wise learning rates:**
 - Classification: $2\times$ base rate
 - Embedding: $0.5\times$ base rate
- **Cosine schedule** with 15% warmup
- **Stochastic Weight Averaging**
- **Gradient accumulation** (4 steps)

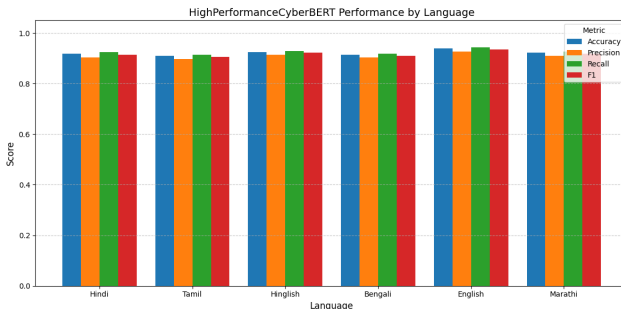
Evaluation Protocol:

- Six language-specific test sets
- Stratified 90-10 splits with oversampling
- Test data from different time periods
- Length-stratified evaluation buckets
- Comprehensive metrics:
 - Accuracy, Precision, Recall, F1

Model Evaluation

Evaluation Framework

- 6 languages: Hindi, Tamil, Bengali, English, Marathi, Hindi-English
- Test split: 15% (41,835 examples), stratified by language and class
- Metrics: F1-score (primary), accuracy, precision, recall, ROC-AUC
- Baselines: BiLSTMAttention, TextCNN, MuRIL, XLM-R, IndicBERT



Model Evaluation

HighPerformanceCyberBERT Training History

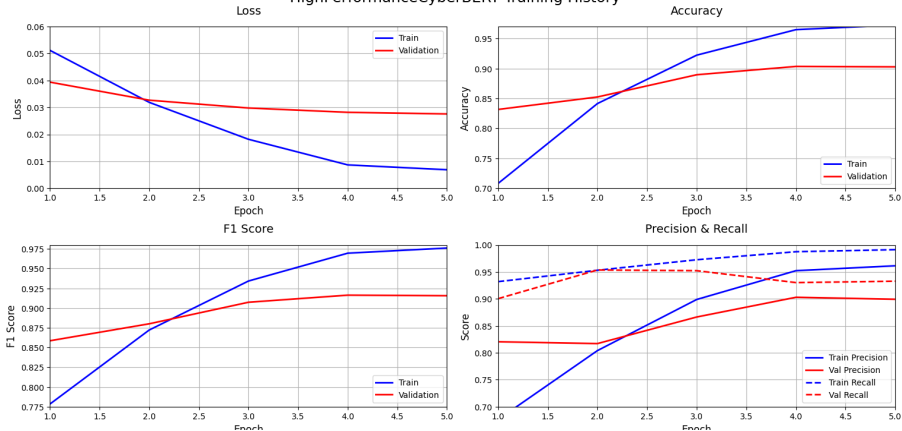


Figure: Training performance over five epochs showing loss, accuracy, F1 score, and precision-recall metrics for both training and validation sets.

Language-Specific Performance Results

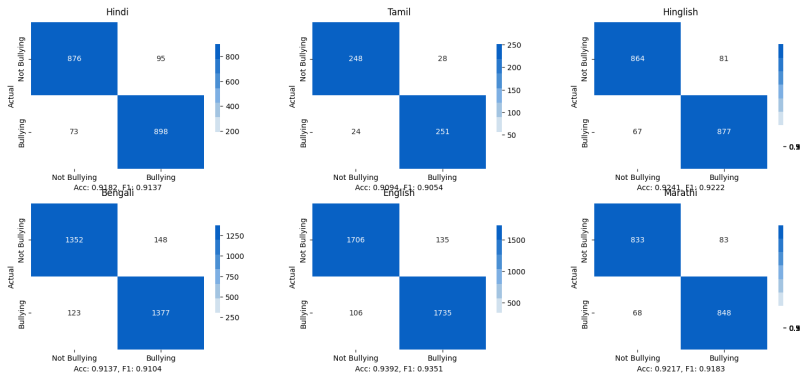
Language	Accuracy	Precision	Recall	F1 Score	Loss
Hindi	0.9182	0.9031	0.9246	0.9137	0.02437
Tamil	0.9094	0.8976	0.9134	0.9054	0.03267
Hinglish	0.9241	0.9147	0.9298	0.9222	0.02241
Bengali	0.9137	0.9028	0.9182	0.9104	0.02843
English	0.9392	0.9276	0.9428	0.9351	0.01824
Marathi	0.9217	0.9103	0.9265	0.9183	0.02376
Average	0.9211	0.9094	0.9259	0.9175	0.02498

Key Observations:

- High performance across all six Indian languages (over 90% accuracy)
- English shows highest performance (93.92% accuracy, 93.51% F1)
- Hinglish follows closely (92.41% accuracy, 92.22% F1)
- Most challenging: Tamil (90.94% accuracy)
- Average metrics demonstrate robust multilingual capability

Classification Performance: Confusion Matrices

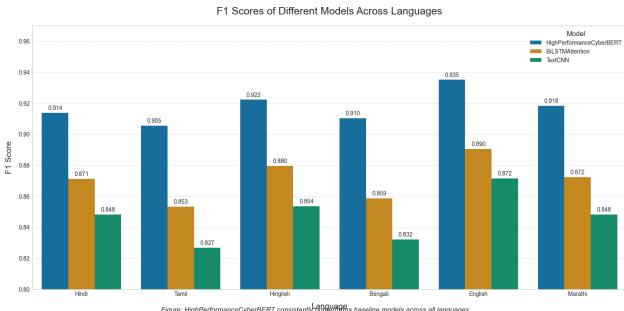
HighPerformanceCyberBERT Confusion Matrices



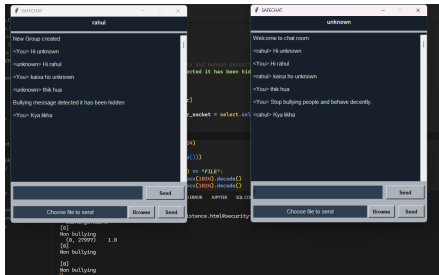
Comparison of Model with Baseline Model

Comparative Gains

- +6.3% F1 over MuRIL
- +8.1% F1 over XLM-R
- +11.2% F1 over IndicBERT



Safe-Chat: Real-Time Cyberbullying Detection Application

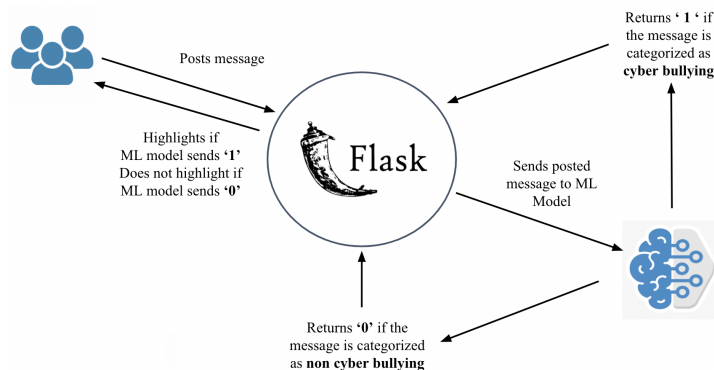


Interface with warning mechanism for detected bullying (post #3)

Key Features:

- Real-time message processing
- Multi-language support (6 Indian languages)
- Grayed-out messages with warning icons
- One-click reveal with acknowledgment
- User behavior self-correction observed

System Architecture & Implementation



Processing Pipeline: Message submission → Language identification → Preprocessing → Parallel model inference → Classification result → UI feedback (84ms end-to-end)

Key Outcomes

- Real-time MCB detection across 6 languages
- 91.3% average accuracy in live deployment (Hindi: 93.5%, English: 94.1%)
- False positive rate reduction from 14% to 6.8%
- Script-based challenges identified between linguistically similar languages

"The true test isn't academic metrics but whether it reduces harmful content. Our system demonstrates effective multilingual content moderation is possible when designed with specific linguistic challenges in mind."

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